Caleb Curran-Velasco

719-644-0767 | calebcurran.velasco@gmail.com | linkedin.com/in/caleb-velasco | calebcurranvelasco.github.io

Education

Colorado School of Mines — Golden, CO
M.S. Computer Science, Machine Learning & AI; GPA: 3.86 — *May 2025*

Colorado School of Mines — Golden, CO
B.S. Computer Science, Cum Laude; GPA: 3.66 — *Aug. 2024*

- Inaugural member of the Presidential Grewcock Scholarship Program (full tuition + fees).
- Relevant Coursework: Differential Equations, Linear Algebra, Discrete Mathematics, Calculus I–III, Probability and Statistics, Advanced Data Science, Data Structures and Algorithms.

Experience

Cloud303 — Denver, CO
Associate Software Engineer (Formerly Intern) — *Jun. 2024 – Present*

- Designed and deployed a real-time ML pipeline to ingest, classify, and semantically search unstructured clinical data (LangChain, Python, Redis, Qdrant, React), enabling real-time insights for a HIPAA-compliant dashboard used by 400+ facilities and 30,000+ patients.
- Built an internal automation platform (NestJS, SvelteKit, AWS SDK, Terraform, RDS) to streamline experimentation and reduce R&D costs, using Git for version control and collaboration.
- Contributed to scalable full-stack development (NestJS, TypeScript, PostgreSQL, MikroORM) and automated deployments (Docker, Terraform, CI/CD), improving reliability and performance.
- Implemented authentication and authorization (Passport, JWT, KeyDB) and optimized containerization/deployment workflows (Docker, Traefik, AWS); created Bash scripts to automatically run inside EC2 instances for reverse proxy configuration and environment setup, improving deployment reproducibility.

Northrop Grumman (via Colorado School of Mines Field Session) — Golden, CO
Student Researcher — *May 2024 – Jun. 2024*

- Engineered a YOLOv8-based vision system with multi-object centroid tracking and heuristic pursuit algorithms in Python, achieving 89% interception accuracy in simulated adversarial scenarios.
- Developed a real-time Unity simulation (C#) with UDP sockets for multi-vehicle communication; collaborated directly with Northrop Grumman engineers to evaluate AI-driven pursuit strategies.

Personal Projects

Breast Cancer Prediction Web App

- Developed a supervised ML model to predict tumor benignity from nuclei measurements; deployed an interactive web app for result visualization.

Real-Time Stock Analysis Tool

- Created a dashboard that tracked top daily stock movers, scraped Yahoo Finance news, and analyzed Twitter sentiment to study correlations with price changes.
- Implemented automated data collection, feature engineering, and visualization pipeline for actionable insights.

Facial Recognition System

- Built a Raspberry Pi-based real-time face detection and recognition system using Python and OpenCV, applying Haar Cascades and LBPH algorithms for accurate identification.
- Integrated webcam streaming and edge inference for low-latency recognition suitable for embedded applications.

Technical Skills

Programming: Python, C/C++, SQL, TypeScript, Java, Bash
Machine Learning: PyTorch, scikit-learn, YOLOv8, NLP, Predictive Modeling, LangChain, Qdrant
Data Engineering: Data Pipelines, ETL, Preprocessing, PostgreSQL
Cloud/DevOps: AWS (EC2, S3, Lambda, EKS, RDS), Docker, Terraform, CI/CD, Traefik, Git
Frameworks: React, NestJS, SvelteKit, MikroORM
Libraries: Pandas, NumPy, Matplotlib, OpenCV
Other: Fluent in English and Spanish; Leadership, Communication, Problem-Solving